

HELIOLITE · REQUIREMENTS

HelioLite requirements

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 2 Technology concept formulated		
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REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with status against the concept estimates
0.1	2026-09-24	Amish Chadha	Initial scaffold

HelioLite requirements

These are first-pass requirements for the concept. Targets are proposals for review, not yet validated with users, and will be checked by calculation at TRL 3. The status column compares each target with the first-order estimates in HLT-PRC-001; "met" means met on paper only.

The **design point** used throughout is a clear winter day at 45° N, DNI 800 W/m², a mirror-to-target distance of 10 m and an angle of incidence on the mirror of about 32° (cosine factor 0.85).

Table 1. Requirements.

ID	Requirement	Target	Status (estimate)	Verification (TRL 3 or later)
R1	Redirected sunlight at the target	200 W or more arriving at the outside of the target glazing at the design point	Met: about 212 W	Optical calculation
R2	Useful daylight in the room	Raise the mean illuminance of a 12 m ² room by 300 lx or more at the design point	Met: about 500 lx (utilization 0.4 assumed)	Illuminance calculation
R3	Daily energy delivered	0.5 kWh or more through the glazing on a clear winter day	Met: about 0.7 kWh	Hourly sun-path calculation
R4	Pointing accuracy	Beam center within 0.5° of the target center (about 90 mm at 10 m), with winds up to 8 m/s	Met on paper: about 0.34° error budget	Error budget; later spot log
R5	Sensorless tracking	Sun position from time and location only; no sun sensor or camera; axis homing switches allowed	Met by design	Design review
R6	Timekeeping	Clock error 60 s or less after 6 months without a network time source	Met: about 32 s (DS3231, ±2 ppm)	Datasheet check
R7	Calibration	One person sets up and calibrates from a phone in 30 min or less, with no survey instrument	Unverified	Procedure review at TRL 3
R8	Working range	Target distance 3 to 20 m; mirror normal azimuth ±135° about the target direction; elevation of the normal -90° (face-down stow) to +90°	Met by design; spill rises above 15 m (see HLT-PRC-001)	Geometry check
R9	Wind	Hold R4 up to 8 m/s; survive 35 m/s (126 km/h) gusts in stow without damage	Not met: the worm gearboxes' holding torque is unknown and may be below the estimated 40 N·m hinge moment if a storm arrives unstowed	Wind load calculation; gearbox data
R10	Safe beam when stowed, faulted or unpowered	Beam directed only at the target or the ground within 3 m of the mast; stow face-down within 60 s of a fault or power loss	Not met: baseline has no stored energy to stow on power loss; a stopped mirror sweeps the beam at about 15°/h	Design review; later fault test
R11	Electrical safety	Only 12 V DC (SELV) outdoors; electronics enclosure IP65; mains adapter indoors and listed	Met by design	Design review
R12	Outdoor life	Operate from -20 to +45 °C; UV-stable plastics (ASA or better); corrosion-resistant mast; 10-year mirror life with cleaning	Unverified	Materials review
R13	Mass and install		Met: about 8 kg on the mast top	Mass estimate

ID	Requirement	Target	Status (estimate)	Verification (TRL 3 or later)
		10 kg or less on the mast top; installed by two people with hand tools in 4 h or less		
R14	Standby power	3 W or less average from the 12 V supply	Met: about 0.6 W	Power budget
R15	Cost	Parts \$400 or less	Met, thin margin: about \$391	Priced BOM (bom/ bom.csv)

■ Requirements not met or at risk

- **R9 (wind survival) not met.** The concept relies on stowing before storms. Without a wind sensor or a network forecast the unit cannot know a storm is coming, and a small NEMA17 worm gearbox may not hold the estimated hinge moment. Options are in HLT-PRC-001.
- **R10 (safe state on power loss) not met.** Worm drives hold the mirror where it stops, and the reflected beam then walks across the surroundings as the sun moves. A small energy store to finish a stow, or a passive cover, is needed.
- **R7 (calibration time) and R12 (outdoor life) unverified.**
- **R15 (cost)** is met with only about \$9 of margin.

■ Assumptions

- Mirror area 0.36 m² (600 x 600 mm), solar-weighted reflectance about 0.93 when clean and 0.95 soiling factor (combined 0.88).
- Double glazing transmits about 75 % of the beam.
- Luminous efficacy of direct sunlight about 95 lm/W; a utilization factor of 0.4 for sunlight bounced from a light ceiling or a diffuser into a 3 x 4 m room.
- Six useful tracking hours on a clear winter day at a mean DNI of 650 W/m² and a mean cosine factor of 0.80.